

HORMONAL REGULATION OF MENSTRUAL CYCLE

- **Regulatory System** : A highly integrated neuroendocrine control system comprising the **Hypothalamus**, **Anterior Pituitary**, and **Ovary** (*Hypothalamo-Pituitary-Ovarian Axis*).
- **Key Hormones Involved** :—
 - a. **Hypothalamic Hormone** : Gonadotropin-Releasing Hormone (GnRH)
 - b. **Anterior Pituitary Hormones** : Follicle-Stimulating Hormone (FSH) & Luteinizing Hormone (LH)
 - c. **Ovarian Hormones** : Estrogen & Progesterone

HYPOTHALAMO-PITUITARY-OVARIAN AXIS

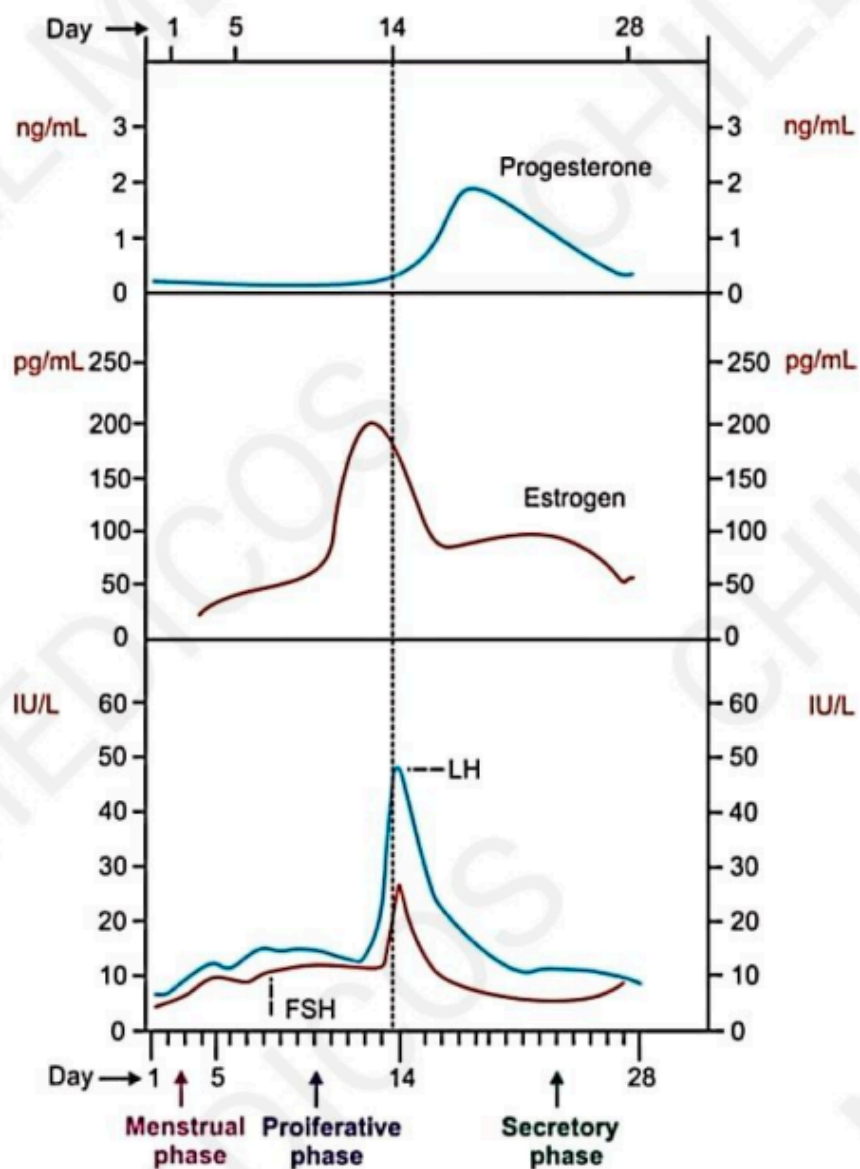
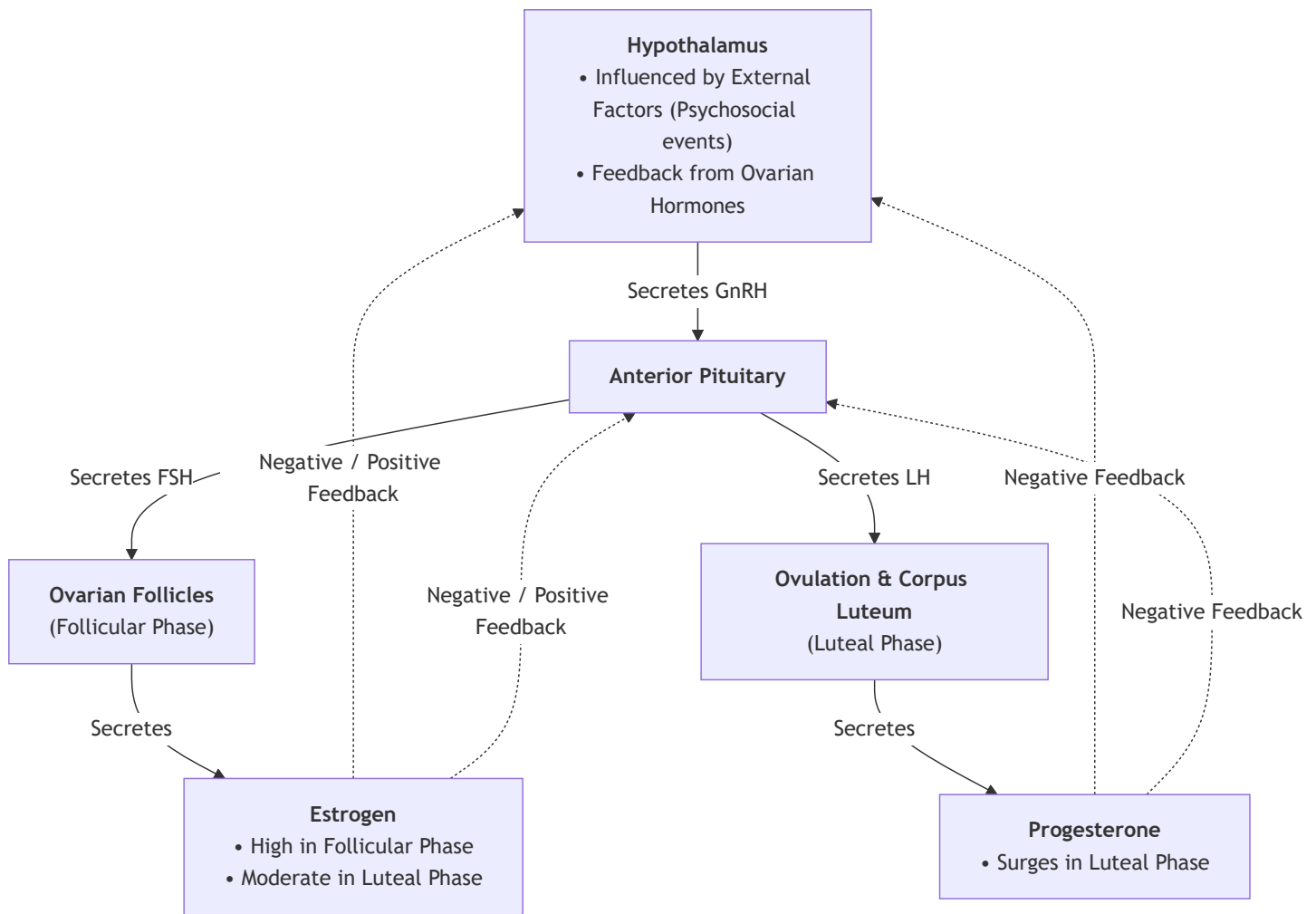


FIGURE 80.4: Hormonal level during menstrual cycle. LH = Luteinizing hormones, FSH = Follicle-stimulating hormone.



1. HYPOTHALAMIC HORMONE — GnRH

- **Mechanism :** GnRH triggers cyclic changes by **stimulating the secretion of FSH and LH** from the anterior pituitary gland.
- **Regulation of Secretion :** GnRH secretion depends upon **2 factors**:
 - a. **External Factors** : Psychosocial events, stress, environmental changes.
 - b. **Feedback Effects** : Feedback regulation exerted by ovarian steroid hormones (estrogen and progesterone).

2. ANTERIOR PITUITARY HORMONES — FSH & LH

- Both **FSH** and **LH** **modulate ovarian and uterine changes** throughout the cycle.
- **Follicle-Stimulating Hormone (FSH) :**

FSH $\longrightarrow$ Stimulates recruitment & growth of immature ovarian follicles

- **Luteinizing Hormone (LH) :**

LH $\longrightarrow$ Triggers ovulation & sustains the corpus luteum

3. OVARIAN HORMONES — ESTROGEN AND PROGESTERONE

- **Estrogen :**

Ovarian Follicles $\longrightarrow$ Estrogen $\longrightarrow$ $\left\{ \begin{array}{l} \text{Follicular Phase} \\ \text{Luteal Phase} \end{array} \right.$

- **Progesterone :**

Corpus Luteum $\longrightarrow$ Progesterone $\longrightarrow$ Luteal Phase

Note : Both ovarian hormones are under the primary regulatory control of **GnRH**, which acts indirectly via **FSH and LH**.